

AquaSentinel AI

A Multi-Agent AI Platform for Smart Water Management

Project Overview

Water is one of our most valuable resources, yet people often face difficulties in understanding water quality, selecting suitable purification methods, conserving water, or reporting water-related issues. These services are usually available on different platforms, making the process confusing and time-consuming.

AquaSentinel AI is a Multi-Agent AI platform developed using n8n to bring these services together through a single conversational interface. Instead of relying on one AI model for every task, the system uses a central AI Manager that intelligently routes each user query to specialized AI advisors. This modular approach allows users to receive focused guidance while keeping the interaction simple and efficient. The project also automates complaint registration, making water-related support easier to access for everyone.

Problem Statement

Water-related problems such as discolored water, bad smell, unsafe drinking water, and supply interruptions are common concerns in many communities. At the same time, people often struggle to choose the right water purifier or understand basic water quality information like TDS. Reporting these issues can also be difficult because complaint systems are often separate from information services.

A single intelligent platform that provides expert guidance while simplifying complaint registration can improve accessibility, reduce confusion, and encourage better public participation in water management.

Proposed Solution

AquaSentinel AI addresses these challenges through a Multi-Agent AI architecture. A central AI Manager first understands the user's request and then delegates it to one or more specialized AI advisors based on the topic.

The platform currently includes dedicated advisors for Water Safety, Water Purification, Water Conservation, and Complaint Registration.

When users need to report an issue, the Complaint Registration Advisor provides access to an integrated complaint form. Submitted complaints are automatically stored in Google Sheets, and users receive a confirmation email. By combining conversational AI with workflow automation, AquaSentinel AI demonstrates how intelligent systems can simplify access to public services while delivering practical and consistent guidance.

Objectives

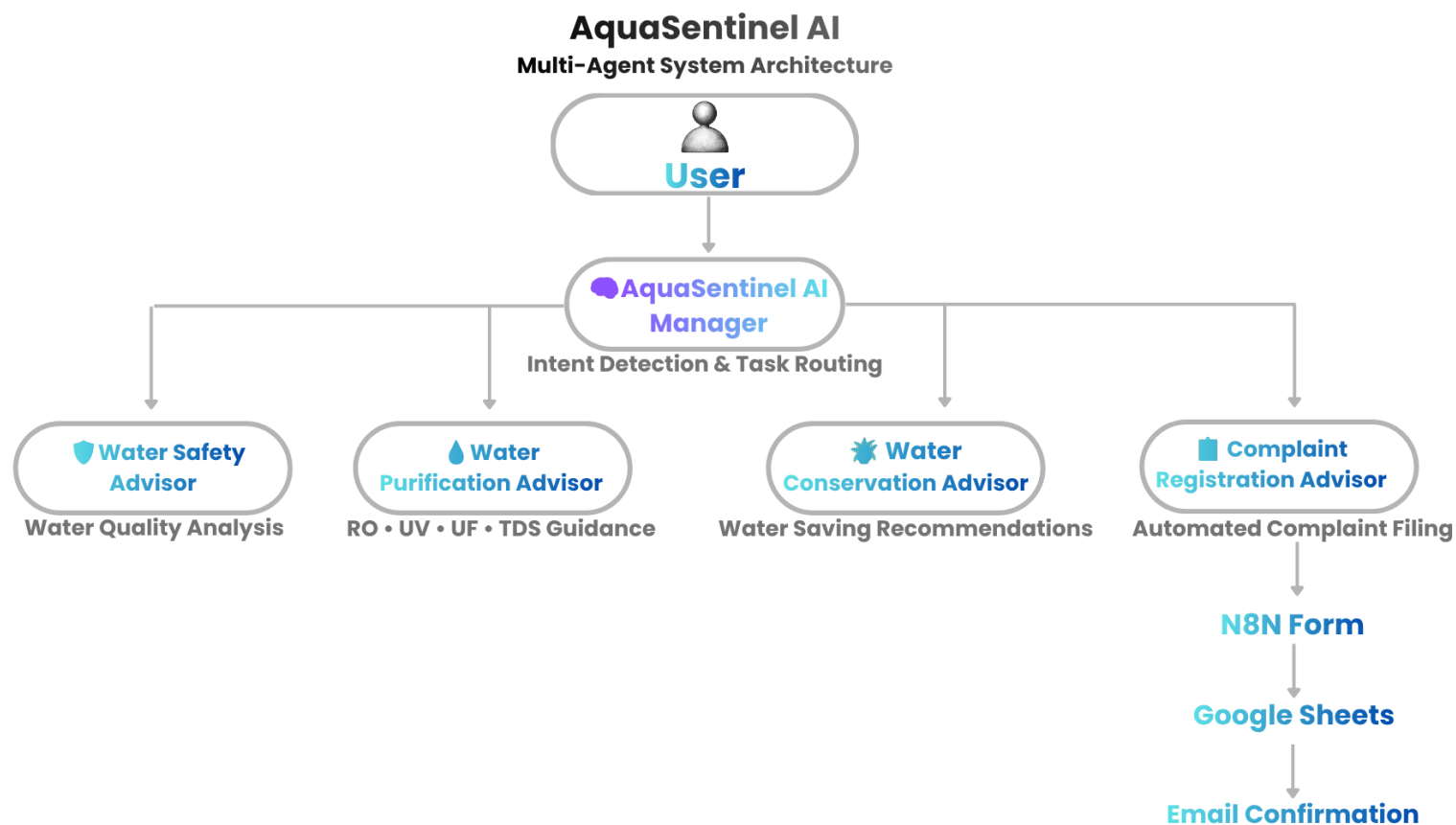
The main objectives of AquaSentinel AI are:

- Provide reliable guidance on water safety, purification, and conservation.
- Simplify water-related complaint registration through automation.
- Promote responsible and sustainable water usage.
- Demonstrate the practical use of Multi-Agent AI in public services.
- Improve user experience by providing a single platform for multiple water-related services.
- Build a scalable foundation for future enhancements such as regional language support, voice interaction, and smart water monitoring.

System Architecture

AquaSentinel AI follows a **Multi-Agent AI architecture** where a central AI Manager acts as the coordinator. It understands the user's request and routes it to the appropriate AI advisor based on the topic. This modular approach allows each advisor to focus on a specific task while the AI Manager combines the responses into a smooth user experience. The complaint registration workflow is further automated using n8n Forms, Google Sheets, and email notifications.

Figure 1: *AquaSentinel AI Multi-Agent System Architecture*



AquaSentinel Multi-Agent AI Architecture

Key Features

- Multi-Agent AI architecture with intelligent task routing.
- Four specialized AI advisors for Water Safety, Purification, Conservation, and Complaint Registration.
- Automated complaint registration with confirmation email.
- Google Sheets integration for complaint records.
- Simple and user-friendly conversational interface.
- Modular workflow built using n8n for easy scalability.

Technology Stack

Component	Technology
Workflow Automation	n8n
AI Models	Gemini 2.5 Flash, Groq Llama
Multi-Agent Coordination	AI Manager
Complaint Form	n8n Form
Database	Google Sheets

Notifications	Gmail
Version Control	GitHub

Expected Impact

AquaSentinel AI aims to make water-related information easier to access while simplifying the complaint registration process. By bringing multiple services into a single platform, it can help users make informed decisions, encourage responsible water usage, and reduce the effort required to report water-related issues. With future development and integration with municipal systems, AquaSentinel AI has the potential to improve access to water-related services, encourage community participation, and support smarter water management initiatives.

Future Scope

The current system serves as a working prototype and can be enhanced further by adding:

- Support for regional languages to reach a wider audience.
- Voice-based interaction for improved accessibility.
- Integration with IoT sensors for real-time water quality monitoring.
- Complaint status tracking through the chatbot.
- Location-based complaint routing.
- Analytics dashboard for authorities to identify recurring water issues.

Conclusion

AquaSentinel AI demonstrates how Multi-Agent AI and workflow automation can be used to solve practical water-related challenges. By combining specialized AI advisors with an automated complaint system, the platform offers a simple and user-friendly solution. With further development, it has the potential to become a valuable digital assistant that supports citizens while contributing to smarter and more accessible public water services.